

Project: Agentic Debate and Persuasion Between Conspiracy and Scientific Models

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Project Output: An end-to-end system that simulates structured debates between two LLM-based agents (a conspiracy proponent and a scientific opponent) moderated by a control agent, runs repeated experiments across multiple conspiracy theories and models, logs all interactions in a unified debate format, and analyzes persuasion dynamics and language use.

Familiarization with LLM Models

The objective of this task is to familiarize with different LLM backends that will later be used as debating agents.

Models / Platforms:

- Gemini (free daily access via API)
<https://aistudio.google.com/api-keys>
- Dockerized Ollama with local models (e.g. LLaMA, Qwen, Mistral)
<https://ollama.com/>
- OpenAI API (discuss with me if needed).

Possible Steps (Adjustable):

- Set up access to Gemini API and run simple prompts
- Install and run Ollama locally via Docker
- For your own understanding compare:
 - response length
 - coherence
 - tendency to hedge or assert
 - Document limitations (rate limits, latency, hallucinations)

Agentic Debate System (Conspiracy vs Scientific Evidence)

The objective is to design and implement a multi-agent debate system in which two agents with opposing epistemic positions debate a conspiracy claim over multiple turns, under the supervision of a moderator agent.

Roles:

- Proponent Agent: Argues in favor of a conspiracy claim provided at the beginning e.g. "COVID-19 vaccines contain microchips".

- Opponent Agent: Argues against the claim using scientific or evidence-based reasoning.
- Moderator Agent: Controls turn-taking, tracks rounds, and decides when the debate ends (either if one of the debating agents change their minds, or after k turns).

The focus is not factual correctness, but persuasion attempts and resistance to persuasion.

Debate Orchestration:

Students may choose one orchestration framework

- CrewAI <https://www.crewai.com/>
- LangGraph
- AutoGen (or equivalent)

Access to Information:

- Each debating agent may use Tavily for online search during the debate.
<https://www.tavily.com/pricing>
- The proponent will search fringe communities and biased queries e.g. "site:[reddit.com/r/conspiracy](https://www.reddit.com/r/conspiracy)".
- The opponent will use mainstream sources e.g. fact-checkers, scientific organizations and reports, etc).

Canonical Debate Log Format: All debates (synthetic or human) must be logged in a common JSON schema, enabling later comparison. This information will be extracted by the moderator.

- debate_id
- claim
- round
- speaker_role (proponent | opponent | moderator)
- utterance
- stance (pro | con | neutral)
- confidence (optional, numeric or categorical)
- tool_used (none | tavily)
- tool_query
- reply_to_turn

Sources:

- <https://blog.gopenai.com/i-built-an-ai-debate-team-to-settle-the-pineapple-on-pizza-argument-32d69291d3cf>
- <https://www.digitalocean.com/community/tutorials/local-ai-agents-with-langgraph-and-ollama>
- <https://microsoft.github.io/autogen/stable//user-guide/core-user-guide/design-patterns/group-chat.html>
- <https://github.com/nikhil-verma-ml/DebateGraph>

Debate Argumentation & Persuasion Analysis

The objective of this task is to analyze argumentation patterns, language use, and persuasion dynamics in debates.

Before synthetic agent debates are available, real debate datasets can be used to set up and validate the analysis pipeline.

- Intelligence Squared Debates (IQ2): <https://tisjune.github.io/research/iq2>
- So you don't have to wait until the output of the debate system, you can use this dataset to process it and analyze it as if it was the dialogue from the agents.
 - design analysis code
 - test metrics
 - establish human persuasion baselines

Data Preparation:

- Convert debates into the same canonical debate log format used by agents
- Assign roles (proponent / opponent / moderator)
- Preserve turn order and debate boundaries

Analysis Objectives: Analyze debates (human and synthetic) with respect to:

- Language use:
 - uncertainty: https://drive.google.com/drive/folders/1vm1jN69BapA36-JbEsYf5zrb9FBfmii0?usp=drive_link
 - moral framing: <https://github.com/medianeuroscience/emfd>
 - modality (strong https://drive.google.com/file/d/1t13LePb_h19Qnez9h5f9qp1u1hRZCxSV/view?usp=drive_link and weak https://drive.google.com/file/d/1r4zSOp-bC7bI0lAnx6lph7ZP52SN0QZV/view?usp=drive_link)
 - emotion <https://saifmohammad.com/WebPages/NRC-Emotion-Lexicon.htm>
- Persuasion outcomes:
 - stance stability or change
 - confidence shifts over turns
- Comparison:
 - human vs LLM debates
 - model vs model (Gemini vs Ollama)